

of Faraday's experimental results. In 1855 and 1856, he proved that electricity and magnetism are actually different aspects of one force: electromagnetism. He then went further—he calculated what the speed of an electromagnetic wave would be if it was produced by an electric current in a conducting loop. The answer was the speed of light. Maxwell realized that light itself must be an electromagnetic wave. Indeed, his equations showed that there could be an infinite number of frequencies of electromagnetic waves, all traveling at the speed of light, and that the light familiar to us ("visible light") is only a small part of this spectrum. Since his time, other kinds of electromagnetic

waves have been discovered, including X-rays, microwaves, and radio waves. Maxwell published his theory in 1864, with a series of equations. After his death, these were summarized into just four, now known as "Maxwell's laws." While the mathematics is not easy to comprehend, its significance cannot be understated.

The theory was a huge conceptual leap in the understanding of the fundamental forces of nature—and it paved the way for Einstein's special theory of relativity. Indeed, Einstein (see pp.198–203) is quoted as saying: "One scientific epoch ended and another began with James Clerk Maxwell."

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